

A stable evaluation method for the pass rush: ELO

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Background

Currently, evaluation metrics for pass blockers and rusher consist of the following:

1. Pass block win rate
2. Aggregated statistics (sacks, hurries, etc.)
3. Film playback

We aim to apply an ELO model to dynamically update player ratings throughout a season, dependent on matchup outcomes

ELO

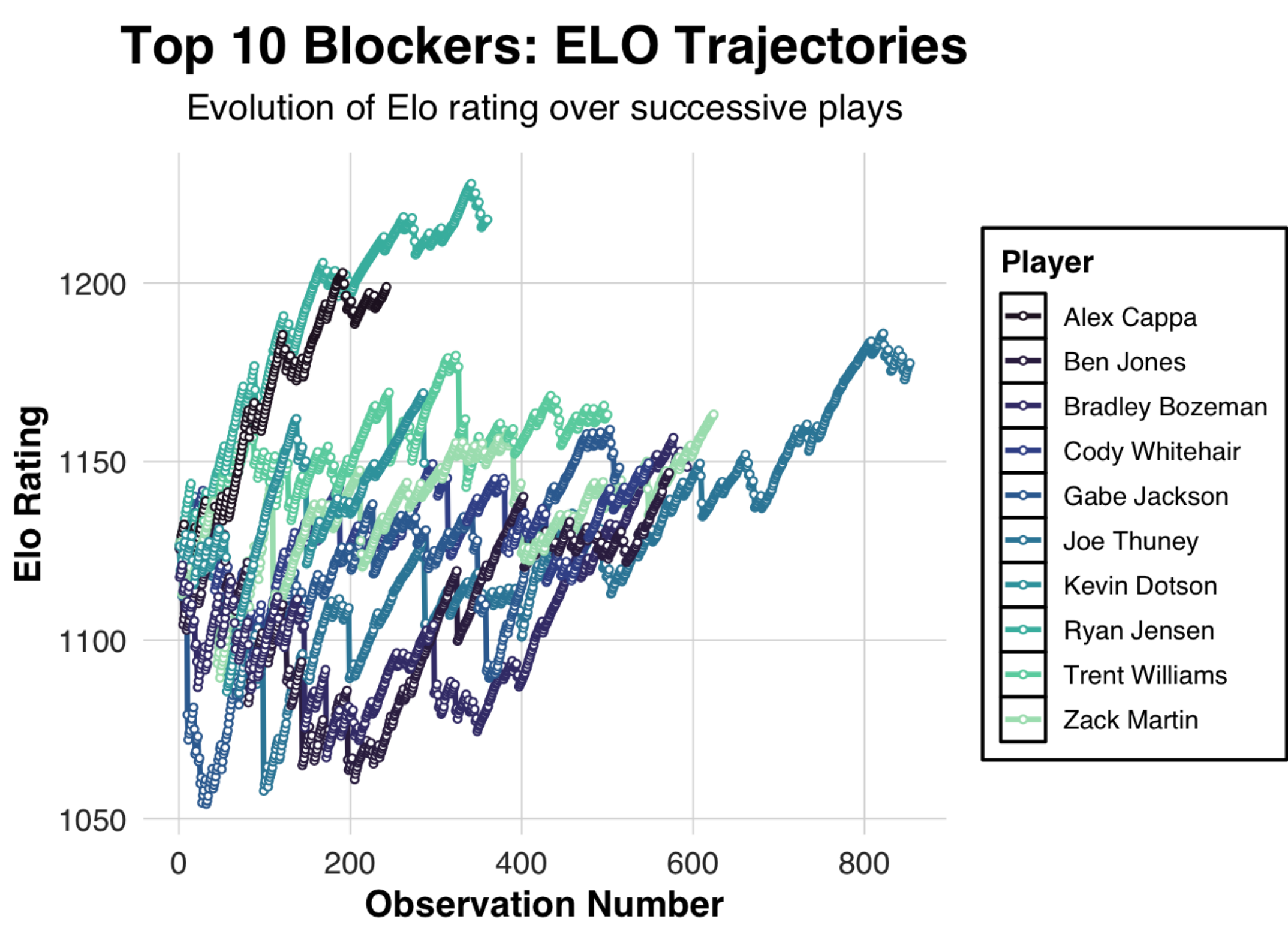
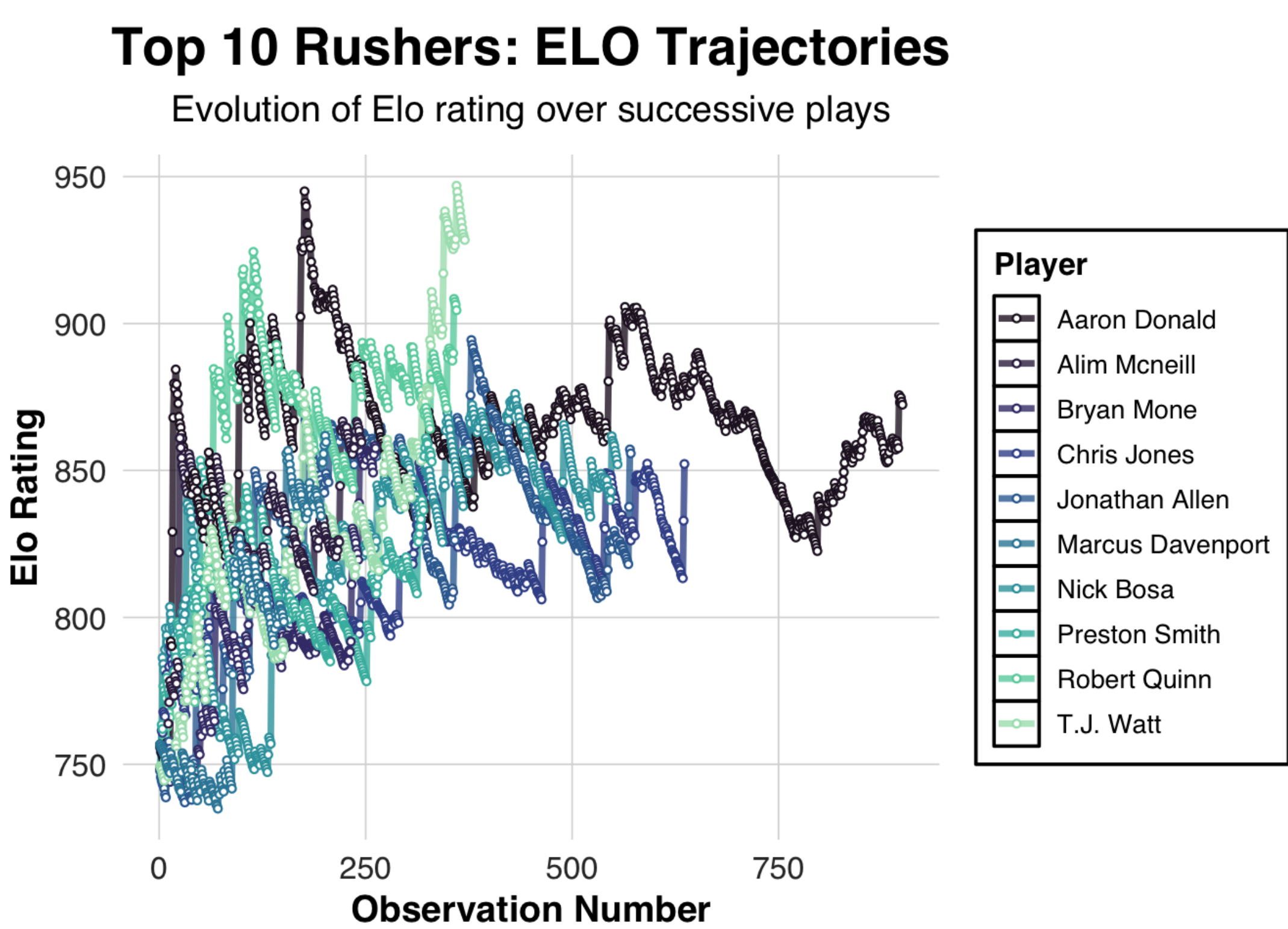
We utilize the ELO rating model to dynamically update player skill levels based on equation 1:

$$R'_A = R_A + K(S_A - E_A)$$

Where R'_A is the new player A ELO, R_A is the original player A ELO, K is the learning rate parameter, S_A is the event outcome, and E_A is the expected event outcome

Results

We apply our model and generate the trajectories of the top 10 players on each side of the ball based on final ending ELO value



Discussion

We validate our model against pure PBWR and a logistic ridge model to predict win/loss of matchups with equation 3.

$$P(A \text{ wins}) = \frac{1}{1 + 10^{(R_B - R_A)/\alpha}}$$

Where $R_B - R_A$ is the ELO difference between matchup players and alpha is our scaling parameter chosen as 319.

Methods

1. Data preparation

Data for the '21-22 season acquired from Hudl was cleaned by filtering for pass plays and grouping by play to add all linemen and quarterback freeze frames for each play

2. Outcome identification

We use Euclidean distance from the QB to identify simple beats, with a weighing of 0.1. We then use tackles on the QB that occur before the pass and behind the line of scrimmage to identify hits, with a weighing of 0.2. Sacks have been pre-identified in the data and are weighed as 1.0

3. ELO updates

We feed each interaction to our multinomial ELO model sequentially. We also utilize an adaptive learning rate, as shown below, to linearly decrease the learning rate after n_{prov} and to set it to a minimum after n_{decay}

$$K(n) = \begin{cases} K_{\text{start}}, & \text{if } n \leq n_{\text{prov}} \\ K_{\text{start}} + (K_{\text{min}} - K_{\text{start}}) \cdot \frac{n - n_{\text{prov}}}{n_{\text{decay}} - n_{\text{prov}}}, & \text{if } n_{\text{prov}} < n \leq n_{\text{decay}} \\ K_{\text{min}}, & \text{if } n > n_{\text{decay}} \end{cases}$$

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